

Azure AI Service Security: Securing with Private Endpoints

Step-by-Step Implementation Guide

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Overview and Prerequisites

What is a Private Endpoint?

A Private Endpoint is a network interface that uses a private IP address within your virtual network (VNet) to connect to an Azure service. It ensures that traffic between your applications and Azure AI Service remains within your private network, eliminating exposure to the public internet.

Benefits of Using Private Endpoints

- **Network Isolation:** Keep traffic within your VNet
- **Enhanced Security:** Eliminates public internet exposure
- **DLP Compliance:** Meets data loss prevention requirements
- **Reduced Attack Surface:** Minimizes vulnerability to threats
- **DNS Integration:** Uses Azure-provided private DNS zones

Prerequisites

Before you begin, ensure you have:

- Active Azure subscription with appropriate permissions
 - Azure AI Services resource (Cognitive Services)
 - Virtual Network (VNet) already configured
 - At least one subnet with available IP addresses
 - Owner or Contributor role in the resource group
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Architecture and Networking Concepts

Architecture Overview

When implementing Private Endpoints for Azure AI Service, the architecture involves:

- **Virtual Network (VNet):** Your private network space
- **Subnet:** A segmented portion of the VNet
- **Private Endpoint:** NIC with private IP in the subnet
- **Private DNS Zone:** Resolves service name to private IP
- **Azure AI Service:** The backend resource

Key Networking Concepts

Private IP Address Range

Private Endpoints use RFC 1918 private IP ranges:

- 10.0.0.0/8
- 172.16.0.0/12
- 192.168.0.0/16

Ensure your VNet and subnet use these ranges to avoid routing conflicts.

DNS Resolution

Create a Private DNS Zone (e.g., [privatelink.cognitiveservices.azure.com](#)) to ensure the service name resolves to the private endpoint's private IP address.

Network Security Groups (NSGs)

Configure NSG rules to allow traffic from your subnet to the private endpoint on port 443 (HTTPS).

Step-by-Step Implementation

Step 1: Create or Identify Your Virtual Network

1. Open the Azure Portal and navigate to **Virtual Networks**
2. Click on your existing VNet or create a new one
3. Note the VNet name and resource group
4. Verify that at least one subnet exists with available IP addresses

Step 2: Locate Your Azure AI Service Resource

1. In the Azure Portal, navigate to **Cognitive Services**
2. Select your Azure AI Service resource
3. Note the resource name and region
4. Click on **Networking** in the left sidebar

Step 3: Create the Private Endpoint

1. In the Networking section, click **Private endpoint connections**

2. Click the **+ Private Endpoint** button

3. Fill in the following details:

Basic Information:

- **Subscription:** Select your Azure subscription
- **Resource group:** Select the resource group for the endpoint
- **Name:** Enter a meaningful name (e.g., **ai-service-pe**)
- **Region:** Select the same region as your Azure AI Service
- Click **Next: Resource**

4. On the **Resource** tab:

- **Connection method:** Select "Connect to an Azure resource in my directory"
- **Subscription:** Select your subscription
- **Resource type:** Select **Microsoft.CognitiveServices/accounts**
- **Resource:** Select your Azure AI Service
- **Target sub-resource:** Select **account**
- Click **Next: Virtual Network**

5. On the **Virtual Network** tab:

- **Virtual network:** Select your VNet
- **Subnet:** Select the subnet where the endpoint will be created
- **Network policy:** Choose "Disable" or "Enable" based on your requirements
- Click **Next: DNS**

6. On the **DNS** tab:

- **Integrate with private DNS zone:** Select "Yes"
- **Private DNS zone:** Select **privatelink.cognitiveservices.azure.com** or create a new one
- Click **Next: Tags**

7. (Optional) Add tags for better resource management

8. Click **Review + Create**

9. Review the configuration and click **Create**

Step 4: Configure DNS Settings

1. Once the Private Endpoint is created, navigate to it
2. Note the **Private IP address** (e.g., 10.0.1.5)
3. Navigate to the Private DNS Zone (**privatelink.cognitiveservices.azure.com**)
4. Verify that an A record exists for your service pointing to the private IP
5. Link the Private DNS Zone to your VNet:
 - Click **Virtual network links** in the DNS zone
 - Click **+ Add**
 - Enter a link name and select your VNet
 - Check **Auto-registration** if desired

- Click **OK**

Step 5: Update Network Security Groups (NSGs)

1. Navigate to the subnet where the private endpoint resides
2. Click on the associated **Network Security Group**
3. Add an inbound rule:
 - **Source:** Your client IP range or VNet address space
 - **Destination:** Private endpoint IP or subnet
 - **Port:** 443 (HTTPS)
 - **Protocol:** TCP
 - **Action:** Allow

Step 6: Disable Public Access (Optional but Recommended)

1. Go back to your Azure AI Service resource
2. Click on **Networking** in the left sidebar
3. Under the **Public access** tab:
 - Select **Disabled** to block all public internet access
 - Save the changes

 **Warning:** Ensure private endpoint is working before disabling public access.

Verification and Testing

DNS Resolution Test

From a VM or client within your VNet, run:

```
nslookup <your-service-name>.cognitiveservices.azure.com
```

Expected result: The output should show the private IP address (e.g., 10.0.1.5) instead of a public IP.

Connectivity Test

Test HTTPS connectivity using curl or Invoke-WebRequest:

```
curl -X GET https://<your-service-name>.cognitiveservices.azure.com/status
```

Or using PowerShell:

```
Invoke-WebRequest -Uri 'https://<your-service-name>.cognitiveservices.azure.com/status'
```

Expected result: HTTP response 200 (success).

Application Integration Test

1. Deploy a test application (e.g., Python, .NET, Node.js) to your VNet that calls the Azure AI Service
 2. Configure the application with the service endpoint URL and API key
 3. Make an API call and verify successful response
 4. Check Azure Activity Logs for successful authentication
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Troubleshooting and Best Practices

Common Issues and Solutions

Issue: DNS Resolution Fails

Cause: Private DNS zone not linked to VNet or wrong DNS configuration.

Solution:

- Verify the private DNS zone is linked to your VNet
- Check DNS servers: Ensure VMs use Azure DNS (168.63.129.16)
- Clear DNS cache and retry

Issue: Connection Timeout

Cause: NSG rules blocking traffic or network routing issues.

Solution:

- Verify NSG inbound rules allow port 443
- Check Network Watcher for connectivity issues
- Verify subnet routes to the private endpoint

Issue: 403 Forbidden Error

Cause: Incorrect API key or authentication failure.

Solution:

- Verify the API key is correct
- Check that the API key has not been regenerated
- Ensure your client is authenticated to Azure
- Check Azure Activity Logs for detailed error information

Best Practices

1. **Use separate Private Endpoints:** Create dedicated endpoints for different environments (dev, test, prod)
2. **Implement NSG rules:** Follow the principle of least privilege
3. **Monitor endpoints:** Set up Azure Monitor alerts for connectivity issues

4. **Document DNS zones:** Keep records of all private DNS zones and linked VNets
 5. **Rotate API keys:** Change API keys regularly for security
 6. **Use managed identities:** Prefer Azure AD authentication over API keys where possible
 7. **Enable public access only for testing:** Disable immediately after verification
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Security Considerations

Network Isolation

By using Private Endpoints, your Azure AI Service communication is isolated within your VNet. This eliminates the need for traffic to traverse the public internet, significantly reducing the attack surface.

Access Control

Combine Private Endpoints with:

- **Azure AD authentication** for identity and access management
- **Role-Based Access Control (RBAC)** for fine-grained permissions
- **Service principals or managed identities** instead of shared API keys

Encryption in Transit

All communication through Private Endpoints uses HTTPS (TLS 1.2+), ensuring encrypted data transmission.

Monitoring and Logging

Enable the following for enhanced security:

- **Azure Monitor:** Track metrics and logs for private endpoint activity
- **Diagnostic Logs:** Capture detailed connection logs
- **Azure Security Center:** Enable continuous security monitoring
- **Activity Logs:** Audit all changes to private endpoints and NSGs

Compliance Requirements

Private Endpoints help meet compliance requirements for:

- **HIPAA:** Healthcare data protection
 - **PCI DSS:** Payment card industry compliance
 - **GDPR:** Data privacy regulations
 - **SOC 2:** Security and controls auditing
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Configuration Checklist

Use this checklist to track your Private Endpoint setup progress:

- ☐ VNet and Subnet Created
- ☐ Private Endpoint Created
- ☐ Private DNS Zone Configured

- ☐ DNS Zone Linked to VNet
 - ☐ NSG Rules Configured
 - ☐ DNS Resolution Tested
 - ☐ Connectivity Verified
 - ☐ Public Access Disabled
 - ☐ Monitoring Enabled
-

Conclusion

Securing your Azure AI Service with Private Endpoints is a critical step in protecting sensitive data and meeting compliance requirements. By following the steps outlined in this guide, you can establish a secure, isolated communication channel between your applications and Azure AI Service.

Remember to:

- Regularly review your network configuration
- Monitor for anomalies
- Update your security posture as your organization evolves
- Refer to the official Microsoft Azure documentation for the latest updates

For additional support and information, visit the [Microsoft Azure Documentation](#) portal.

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Additional Resources

Microsoft Documentation

- [Azure Private Endpoints Documentation](#)
- [Azure Cognitive Services](#)
- [Virtual Networks and Subnets](#)
- [Private DNS Zones](#)

Related Topics

- Network Security Groups (NSGs)
 - Azure Virtual Network
 - Azure Private Link
 - Azure Firewall
 - Application Gateway with Private Endpoints
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FAQ

Q: Can I have multiple private endpoints for the same Azure AI Service?

A: Yes, you can create multiple private endpoints for the same service in different VNets or subnets for high availability and redundancy.

Q: What happens to my existing public endpoint when I create a private endpoint?

A: The public endpoint remains accessible until you explicitly disable it. You can have both active simultaneously during testing.

Q: How do I test connectivity from outside my VNet?

A: Use Azure Bastion or a VPN connection to access a VM within your VNet, then test from there.

Q: Can I use private endpoints with managed identities?

A: Yes, managed identities work seamlessly with private endpoints and are the recommended approach.

Q: What is the cost of using Private Endpoints?

A: Private Endpoints have a per-hour charge. Check Azure pricing page for current rates.

Q: Can I delete a private endpoint without affecting the Azure AI Service?

A: Yes, deleting a private endpoint only removes the private connection path. The service remains functional with other endpoints.

Q: How do I monitor private endpoint activity?

A: Use Azure Monitor, Diagnostic Logs, and Activity Logs to track all connectivity and configuration changes.

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